

SETAREH NAJAFI

✉ setareh@ucsb.edu  setareh-najafi  setarehnj  setarehnj.github.io ☎ (805) 858-0902

EDUCATION

Ph.D. in Computer Science, University of California, Santa Barbara *GPA: 4.0/4.0* Jan. 2023 – 2028
M.S. in Computer Science, University of California, Santa Barbara *GPA: 4.0/4.0* Jan. 2023 – Sep. 2025
B.Sc. in Computer Science, University of Tehran *GPA: 3.86/4.0* Sep. 2019 – July 2022

Mathematical Coursework: Abstract Algebra, Linear Algebra, Probability, Stochastic Processes, Statistical Methods, Combinatorics, Graph Theory, Number Theory, Mathematical Logic, Theory of Computation, Game Theory, Scientific Computing

RESEARCH INTERESTS

Mechanistic Interpretability
Chain-of-Thought Reasoning

Machine Unlearning in LLMs
Reinforcement Learning

RESEARCH EXPERIENCE

Independent AI Safety Researcher Mar. 2026 – Present

- Investigating language-model safety, with a focus on chain-of-thought faithfulness, mechanistic interpretability, and robustness to adversarial prompting.
- Investigating non-verbalized evaluation awareness in language models

Research Assistant, UCSB, Advisor: Prof. Z. Zhang Mar. 2024 – Present

- Worked on Machine Unlearning for Large Language Models
- Worked on Kolmogorov-Arnold Networks (KANs) for reachability analysis
- Implemented post-training compression of Llama 3B using SVD-based low-rank approximation
- Developed DeepONet-based preconditioning to accelerate GMRES for 3D-IC thermal simulation

Research Assistant, UCSB, Advisor: Prof. E. Vigoda Jan. 2023 – April 2023

- Worked on designing parallel and distributed MCMC algorithms for random proper vertex colorings in general graphs

Research Assistant, University of Tehran, Advisor: Prof. A. Ghadermarzi Feb. 2022 – Aug. 2022

- Studied elliptic curves over finite fields and their applications to public-key cryptography
- Analyzed ECDH, ElGamal, digital signatures, and the elliptic curve discrete logarithm problem

Research Assistant, University of Tehran, Advisor: Prof. B. Bahrak Aug. 2021 – Jan. 2022

- Worked on developing a stochastic model for large-scale networks where traditional MCMC methods are less effective using stochastic processes

MASTER'S RESEARCH PROJECT

DeepONet-Based Preconditioning for 3D-IC Thermal Simulation 2025

- Developed a hybrid multigrid preconditioner using DeepONet trunk basis functions to accelerate GMRES convergence for 3D-IC thermal simulation

GRADUATE COURSE RESEARCH PROJECTS

Machine Learning On Graphs Fall 2023

- Image reconstruction from fMRI data Using Graph Representation and Diffusion Models

Information Retrieval and Web Search Fall 2023

- Adapting a Discriminative Text Encoder for Dense Retrieval via Contrastive Span Prediction for QA Datasets

Automated Verification Spring 2023

- Modeled and verified WireGuard security properties using Tamarin Prover

Software Engineering Spring 2024

- Integrated AFL++ fuzzing into a GitHub Actions CI/CD pipeline for automated vulnerability detection

Advanced Topics in Security Fall 2023

- Graph Neural Network Approach to Classifying Vulnerabilities in Binary Files via Control Flow Graph Analysis

SELECTED PROJECTS

Deep Generative Models

- Implemented a GAN, DCGAN, and WGAN model to reproduce hand-written digits using MNIST dataset
- Implemented a convolutional autoencoder to reproduce color images using CIFAR-10 dataset

Language/Computer Vision Models

- Implemented an object detection model by developing non-max suppression on anchor boxes in YOLO model
- Implemented a sentence classification model using GloVe embedding and LSTM layers for adding emojis to sentences

Grapevine Leaves Classification

- Implemented multiple deep machine learning models for grapevine leaf classification using a transfer learning approach based on pre trained CNNs, namely MobileNetV2, ResNet50, EfficientNetB3, and InceptionNetV3
- Implemented an autoencoder to improve model accuracy and used 10 fold cross validation for evaluation

LinkedIn Implementation

- Developed an object-oriented program that had LinkedIn functionalities with graphical interface

Classroom Object Oriented Language(COOL) Compiler

- Implemented a compiler in C++ for the Classroom Object-Oriented Language(COOL), including a syntax analyzer using Flex, a parser using Bison, and a semantic analyzer using logic inference rules

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, MATLAB, HTML

Machine Learning: PyTorch, PyTorch Lightning, TensorFlow, Keras, scikit-learn

LLMs & Agents: Hugging Face Transformers, TransformerLens, LangChain, LangGraph, OpenAI API, Anthropic API, vLLM

Scientific Computing: NumPy, pandas, SciPy

Engineering Tools: CUDA, Git, Linux, Docker, GitHub Actions, AWS

Databases: MySQL

Security & Formal Verification: Tamarin Prover, AFL++, Z3

TEACHING EXPERIENCE

Teaching Assistant, University of California, Santa Barbara

Advanced Topics in Object-Oriented Computing	Winter 2025
Problem Solving with Computers	Fall 2024
Foundations of Computer Science	Winter & Spring 2024

Teaching Assistant, University of Tehran

Engineering Probability and Statistics	Fall 2021
Number Theory	Fall 2021
Linear Algebra	Fall 2020
Fundamentals of Combinatorics	Fall 2020

HONORS AND AWARDS

- Computer Science Department Summer Fellowship Award, University of California, Santa Barbara, Summer 2024
- University of California Santa Barbara Ph.D. Fellowship and Academic Excellence Award
- NYU Tandon School of Engineering Ph.D. Fellowship Offer
- Ranked 280 among more than 300000 participants in Iranian Nationwide Physics and Mathematics University Entrance Exam
- Semi-finalist in National Olympiad of Mathematics and Informatics
- Accepted in Iranian National Organization for Development of Exceptional Talents High Schools